Assignment 2

Convex Optimisation

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Question 1 (Maximum Entropy with Moment Constraints)

We have a set of n k-dimensional vectors $\{\phi_i\}_{i=1}^n \subset \mathbb{R}^k$. We wish to solve the following constrained optimization problem:

$$\min_{P \in \mathbb{R}^n} \quad H_{\text{neg}}(P) = \sum_{i=1}^{n} p_i \log p_i \tag{1}$$

I have replaced $H(P)$ by its negative to make the objective function of the optimisation problem convex in P. The following constraints must be satisfied :

$$\sum_{i=1}^{n} p_i \phi_i = \mu, \tag{2}$$

$$P = (p_1, \ldots, p_n) \in \Delta_n, \tag{3}$$

where:

1. $\mu \in \mathbb{R}^k$ is the target vector,
2. Δ_n (the probability simplex) is defined as

$$\Delta_n = \left\{ P \in \mathbb{R}_+^n : \sum_{i=1}^{n} p_i = 1 \right\}. \tag{4}$$

(a) We derive the dual function and the dual optimisation problem in this sub part.

1) We use the dual variables

$$\theta = \{\theta_1, \theta_2, \ldots, \theta_k\}$$

for the moment constraint given in equation (2).

hence

$$\theta \in \mathbb{R}^k.$$

2) We use the dual variables

$$\nu = \{\nu_1, \nu_2, \ldots, \nu_n, \nu_{n+1}\}$$

for the simplex constraint given in equation (3), where:

1. We use $\nu_1, \nu_2, \ldots, \nu_n$ for the inequality constraints

$$-p_i \leq 0 \quad \text{for } i = 1, 2, \ldots, n.$$

Therefore,

$$\nu_i \geq 0 \quad \text{for } i = 1, 2, \ldots, n.$$

2. We use ν_{n+1} for the equality constraint

$$\sum_{i=1}^{n} p_i - 1 = 0,$$

hence

$$\nu_{n+1} \in \mathbb{R}.$$

We construct the Lagrangian as follows:

$$\mathcal{L}(P, \theta, \nu) = \sum_{i=1}^{n} p_i \log(p_i) + \sum_{j=1}^{k} \theta_j \left(\sum_{i=1}^{n} p_i \Phi_{ij} - \mu_j \right) - \sum_{i=1}^{n} \nu_i p_i + \nu_{n+1} \left(\sum_{i=1}^{n} p_i - 1 \right), \tag{5}$$

where:

1) Φ_{ij} is the (i,j)-th element of the matrix $\Phi \in \mathbb{R}^{n\times k}$, which is constructed by collecting the vectors $\{\phi_i\}_{i=1}^n$ and stacking them as rows.

Simplifying (5), we get:

$$\mathcal{L}(P,\theta,\nu) = \sum_{i=1}^{n} p_i \log(p_i) + \sum_{i=1}^{n}\sum_{j=1}^{k} p_i \Phi_{ij} \theta_j - \sum_{j=1}^{k} \theta_j \mu_j - \sum_{i=1}^{n} \nu_i p_i + \nu_{n+1}\left(\sum_{i=1}^{n} p_i - 1\right). \tag{6}$$

The dual function $g(\theta,\nu)$ is given by

$$g(\theta,\nu) = \inf_{P\in\mathbb{R}^n}\left[\sum_{i=1}^{n} p_i \log(p_i) + \sum_{j=1}^{k}\theta_j\left(\sum_{i=1}^{n} p_i \Phi_{ij} - \mu_j\right) - \sum_{i=1}^{n}\nu_i p_i + \nu_{n+1}\left(\sum_{i=1}^{n} p_i - 1\right)\right]. \tag{7}$$

The function on the RHS of the above equation is convex in P. Hence, its minimum with respect to P will be achieved at the point which satisfies:

$$\nabla_P \mathcal{L}(P,\theta,\nu) = 0. \tag{8}$$

To find the P which minimizes the RHS of the equation, we do the following:

The i-th element of $\nabla_P \mathcal{L}(P,\theta,\nu)$ is given by

$$\frac{\partial \mathcal{L}(P,\theta,\nu)}{\partial p_i} = 1 + \log(p_i) + \sum_{j=1}^{k}\Phi_{ij}\theta_j - \nu_i + \nu_{n+1}. \tag{9}$$

For the optimality condition to be satisfied, we require

$$\nabla_P \mathcal{L}(P,\theta,\nu) = 0 \quad \text{for } i = 1,2,\ldots,n, \tag{10}$$

which translates to

$$1 + \log(p_i) + \sum_{j=1}^{k}\Phi_{ij}\theta_j - \nu_i + \nu_{n+1} = 0 \quad \text{for } i = 1,2,\ldots,n. \tag{11}$$

Solving for p_i, we obtain:

$$p_i = \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k}\Phi_{ij}\theta_j\right), \quad \text{for } i = 1,2,\ldots,n. \tag{12}$$

Substituting equation 12 in equation 6, we get

Substituting the expression for p_i into the dual function, we get

$$\begin{aligned} g(\theta,\nu) = &\sum_{i=1}^{n} \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right)\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right) \\ &+ \sum_{j=1}^{k} \theta_j \sum_{i=1}^{n} \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{l=1}^{k} \Phi_{il}\theta_l\right)\Phi_{ij} - \sum_{j=1}^{k} \theta_j \mu_j \\ &- \sum_{i=1}^{n} \nu_i \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right) \\ &+ \nu_{n+1}\left(\sum_{i=1}^{n} \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right) - 1\right). \end{aligned} \tag{13}$$

Simplifying the above equation, we obtain

$$g(\theta,\nu) = -\sum_{i=1}^{n} \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right) - \sum_{j=1}^{k} \theta_j \mu_j - \nu_{n+1}. \tag{14}$$

The dual optimization problem can be formulated using the dual function $g(\theta,\nu)$ as follows.

Dual Problem:

$$\sup_{\substack{\theta\in\mathbb{R}^k \\ \nu\in\mathbb{R}^{n+1}}} \quad g(\theta,\nu) \tag{15}$$

subject to

$$\nu_i \geq 0 \quad \text{for } i = 1, 2, \ldots, n. \tag{16}$$

(b) The conditions under which strong duality holds with respect to (Φ, μ) are given as follows:

1) The Primal optimisation problem must be a Convex Optimisation problem.

2) There must exist $P \in \mathbb{R}^n$ which satisfies the equality constraints of the problem, and strictly satisfies the inequality constraints.

That is, there should exist

$$P = \{p_1, p_2, \ldots, p_n\} \tag{17}$$

such that:

1.
$$\sum_{i=1}^{n} p_i \phi_i = \mu \tag{18}$$

2.
$$\sum_{i=1}^{n} p_i = 1 \tag{19}$$

3.
$$p_i > 0 \quad \text{for } i = 1, 2, \ldots, n \tag{20}$$

3) The optimal value of the dual problem defined in equation 15, denoted by $d^\star$, must satisfy

$$d^\star > -\infty. \tag{21}$$

where

$$d^\star = \sup_{\substack{\theta\in\mathbb{R}^k \\ \nu\in\mathbb{R}^{n+1}}} g(\theta,\nu), \qquad \text{with } \nu_i \geq 0 \ \text{ for } i = 1, 2, \ldots, n. \tag{22}$$

Regarding the conditions for strong duality listed above

Condition **1)** is satisfied for all Φ and μ as

Consider the function

$$f(P) = \sum_{i=1}^{n} p_i \log(p_i). \tag{23}$$

The function $f(P)$ is convex in P. This can be shown using the fact that the Hessian of $f(P)$ is given by

$$\nabla^2 f(P) = \begin{bmatrix} \frac{1}{p_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{p_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{p_n} \end{bmatrix}. \tag{24}$$

Since

$$p_i > 0 \quad \text{for } i = 1, 2, \ldots, n \tag{25}$$

(due to the presence of $\log(p_i)$ in $f(P)$), each diagonal entry of $\nabla^2 f(P)$ is positive.

Hence,

$$\nabla^2 f(P) \succeq 0, \tag{26}$$

which implies that $f(P)$ is convex.

Also, the constraints in P in the primal optimisation problem given by equations 2 and 3 are all linear in P, and hence they form a convex set. Hence, the primal problem is a Convex Optimisation problem for all $\Phi \in R^{m \times n}$ and $\mu \in R^m$.

The satisfaction of Condition **2)** in the above mentioned conditions depends on the choice of Φ and μ.

Checking the obeyance of Condition **3)** with respect to Φ and μ :

Looking at the dependence of $g(\theta, \nu)$ on $\{\nu_1, \nu_2, \ldots, \nu_n\}$, we observe that

$$\arg\sup_{\nu_i} \; g(\theta, \nu) = 0 \qquad \text{for } i = 1, 2, \ldots, n. \tag{27}$$

Hence, the dual problem in (15) reduces to

$$\sup_{\substack{\theta \in \mathbb{R}^k \\ \nu_{n+1} \in \mathbb{R}}} \left[-\sum_{i=1}^{n} \exp\left(-1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij} \theta_j \right) - \sum_{j=1}^{k} \theta_j \mu_j - \nu_{n+1} \right]. \tag{28}$$

Substituting

$$\nu_{n+1} = 0, \qquad \theta_j = 0 \;\text{ for } j = 1, 2, \ldots, k,$$

into 28, we obtain a lower bound for $d^\star$ defined in equation 22.

Hence, we get

$$d^\star \geq -\sum_{i=1}^{n} \exp(-1), \tag{29}$$

which can be written as

$$d^\star \geq -\frac{n}{e}. \tag{30}$$

Hence, the condition in equation 21 is satisfied for all finite Φ and μ.

Therefore, for finite Φ and μ, if there exits a $P \in R^n$ such that the conditions in equations 17, 18, 19 and 20 are satisfied, then strong duality holds.

(c) We wish to determine the form of the primal maximiser p^* (minimiser for the primal in 1) and examine some of its properties.

In equation (12), we derived a relation between p^*, θ^* and v^*, where p^*, θ^* and v^* are the argument values for the Lagrangian $\mathcal{L}(p, \theta, v)$ when it attains its optimal value for the dual optimisation problem.

The dual problem is given by equations 15 and 16.

We previously explained the conditions under which strong duality holds in part **(b)** of this question. Assuming these conditions are satisfied, strong duality implies that the optimal value of the primal problem equals the optimal value of the dual problem.

Moreover, the p^* that solves the primal problem is the same p^* that appears as part of the optimal solution to the dual problem in equations 15 and 16.

Hence, the relation in equation (12) is to be interpreted as connecting the primal optimiser p^* with the dual optimisers θ^* and v^*.

Restating this we have the expression for the primal optimiser p_i^* given by

$$p_i^* = \exp\left(v_i^* - 1 - v_{n+1}^* - \sum_{j=1}^{k} \Phi_{ij}\, \theta_j^* \right), \qquad \text{for } i = 1, 2, \ldots, n. \tag{31}$$

Here,

$$p^* = \{p_1^*, p_2^*, \ldots, p_n^*\} \tag{32}$$

is the primal optimiser.

Further,

$$v^* = \{v_1^*, v_2^*, \ldots, v_{n+1}^*\} \tag{33}$$

and

$$\theta^* = \{\theta_1^*, \theta_2^*, \ldots, \theta_k^*\} \tag{34}$$

are the dual optimisers.

Now we move on to derive v^* and θ^*, which is the point that optimises the dual problem.
The dual problem is given by

$$\sup_{\theta \in \mathbb{R}^k,\, v \in \mathbb{R}^{n+1}} \left[-\sum_{i=1}^{n} \exp\left(v_i - 1 - v_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\, \theta_j \right) - \sum_{j=1}^{k} \theta_j\, y_j - v_{n+1} \right], \tag{35}$$

subject to

$$v_i \geq 0 \qquad \text{for } i = 1, 2, \ldots, n. \tag{36}$$

Looking at equation (35), it is clear that

$$\underset{v_i \geq 0,\ i=1,\ldots,n}{\arg\max} \; g(\theta, v) = 0. \tag{37}$$

where the dual function $g(\theta, v)$ is defined in equation (14).

Hence, the dual problem now reduces to

$$\sup_{\theta \in \mathbb{R}^k,\ v_{n+1} \in \mathbb{R}} \left[-\sum_{i=1}^{n} \exp\left(-1 - v_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\, \theta_j \right) - \sum_{j=1}^{k} \theta_j y_j - v_{n+1} \right]. \tag{38}$$

If we now replace the objective function in equation (38) by its negative and instead of taking a supremum we take an infimum, we consider the equivalent problem

$$\inf_{\theta\in\mathbb{R}^k,\, v_{n+1}\in\mathbb{R}} \left[\sum_{i=1}^{n} \exp\left(-1 - v_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\, \theta_j \right) + \sum_{j=1}^{k} \theta_j y_j + v_{n+1} \right]. \tag{39}$$

The values of θ and v_{n+1} that solve equations (38) and (39) are the same.

Hence, we determine θ^* and v^*_{n+1} by solving the problem in equation (39).

The objective function in equation (39) is convex in v_{n+1} and θ. Therefore, we equate partial derivatives to 0 to obtain v^*_{n+1} and θ^*.

Hence, v^*_{n+1} and θ^* must satisfy:

1) Derivative with respect to $v_{n+1} = 0$:

$$1 - e^{-1-v_{n+1}} \sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta_j \right) = 0. \tag{40}$$

Rearranging (40), we obtain

$$e^{v^*_{n+1}} = e^{-1} \sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right). \tag{41}$$

2) Derivative with respect to $\theta_\alpha = 0$ for $\alpha = 1, 2, \ldots, k$:

$$e^{-1-v^*_{n+1}} \sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right) (\Phi_{i\alpha}) + \mu_\alpha = 0. \tag{42}$$

Which is simplified to

$$\mu_\alpha\, e^{1+v^*_{n+1}} = \sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right) (\Phi_{i\alpha}), \qquad \alpha = 1, 2, \ldots, k. \tag{43}$$

Equation (43) can be rearranged using equation 41 as

$$\mu_\alpha = \frac{\sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right) \Phi_{i\alpha}}{\sum_{i=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right)}, \qquad \alpha = 1, 2, \ldots, k. \tag{44}$$

Now coming to the determination of the primal solution, we have

$$p^*_i = \exp\left(v^*_i - 1 - v^*_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right), \qquad \text{for } i = 1, 2, \ldots, n. \tag{45}$$

Substituting $v^*_i = 0$ for $i = 1, 2, \ldots, n$ and using the expression for $\exp(-1 - v^*_{n+1})$ from equation (41), we obtain

$$p^*_i = \frac{\exp\left(- \sum_{j=1}^{k} \Phi_{ij}\, \theta^*_j \right)}{\sum_{l=1}^{n} \exp\left(- \sum_{j=1}^{k} \Phi_{lj}\, \theta^*_j \right)}, \qquad \text{for } i = 1, 2, \ldots, n. \tag{46}$$

Now substituting

$$\sum_{j=1}^{k} \Phi_{ij}\, \theta_j^* = (\theta^*)^\top \Phi_i, \tag{47}$$

into equation (46), we obtain

$$p_i^* = \frac{\exp\big(-(\theta^*)^\top \Phi_i\big)}{\sum_{l=1}^{n} \exp(-(\theta^*)^\top \Phi_l)}, \qquad \text{for } i = 1, 2, \ldots, n. \tag{48}$$

We notice that equation (48) implies the proportional form

$$p_i^* \propto \exp\big(-(\theta^*)^\top \Phi_i\big)\,. \tag{49}$$

In place of:

$$p_i^* \propto \exp\big((\theta^*)^\top \Phi_i\big)\,. \tag{50}$$

as given in the question.

This is because θ is the dual variable corresponding to the equality constraint

$$\sum_{i=1}^{n} p_i\, \Phi_i = \mu, \tag{51}$$

which can equivalently be written as

$$\sum_{i=1}^{n} p_i\, \Phi_i - \mu = 0, \tag{52}$$

or

$$\mu - \sum_{i=1}^{n} p_i\, \Phi_i = 0. \tag{53}$$

The way we treat this constraint determines the sign of θ^*.

Finally, from equation (46), we can see that

$$\sum_{i=1}^{n} p_i^* = 1. \tag{54}$$

Now we evaluate

$$\sum_{i=1}^{n} p_i^*\, \Phi_i = \frac{\sum_{i=1}^{n} \exp\big(-(\theta^*)^\top \Phi_i\big)\, \Phi_i}{\sum_{i=1}^{n} \exp(-(\theta^*)^\top \Phi_i)}. \tag{55}$$

where Φ_i denotes the i-th row of the matrix $\Phi \in \mathbb{R}^{n \times k}$

The α-th element of $\displaystyle\sum_{i=1}^{n} p_i^*\, \Phi_i$ is

$$\left(\sum_{i=1}^{n} p_i^*\, \Phi_i\right)_\alpha = \frac{\sum_{i=1}^{n} \exp\big(-(\theta^*)^\top \Phi_i\big)\ \Phi_{i\alpha}}{\sum_{i=1}^{n} \exp(-(\theta^*)^\top \Phi_i)}, \qquad \alpha = 1, 2, \ldots, k. \tag{56}$$

Using equation (44), we can say that

$$\left(\sum_{i=1}^{n} p_i^*\, \Phi_i\right)_\alpha = \mu\alpha. \tag{57}$$

Therefore,

$$\sum_{i=1}^{n} p_i^*\, \Phi_i = \mu. \tag{58}$$

(d) The dual function, as in equation 14, is given by

$$g(\theta,\nu) = -\sum_{i=1}^{n} \exp\left(\nu_i - 1 - \nu_{n+1} - \sum_{j=1}^{k} \Phi_{ij}\theta_j\right) - \sum_{j=1}^{k} \theta_j \mu_j - \nu_{n+1}. \tag{59}$$

The dual problem in equations 15 and 16 involves maximising $g(\theta,\nu)$ over ν such that

$$\nu_i \geq 0 \qquad \text{for } i = 1, 2, \ldots, n.$$

After maximising $g(\theta,\nu)$ over $\nu \in \mathbb{R}^{n+1}$ which satisfy the required constraints, we can write the dual function $g(\theta,\nu^*)$ solely in terms of θ.

The optimiser ν^* satisfies the following:

$$\nu_i^* = 0 \qquad \text{for } i = 1, 2, \ldots, n,$$

and

$$e^{\,\nu_{n+1}^*} = e^{-1} \sum_{i=1}^{n} \exp\left(-\sum_{j=1}^{k} \Phi_{ij}\,\theta_j\right),$$

as given by equations 37 and 41.

Substituting these values of ν^* into equation (14), we get

$$g(\theta,\nu^*(\theta)) = -\frac{\displaystyle\sum_{i=1}^{n} \exp\left(-\sum_{j=1}^{k} \Phi_{ij}\theta_j\right)}{\displaystyle\sum_{i=1}^{n} \exp\left(-\sum_{j=1}^{k} \Phi_{ij}\theta_j\right)} - \sum_{j=1}^{k} \theta_j \mu_j + 1 - \ln\left(\sum_{i=1}^{n} \exp\left(-\sum_{j=1}^{k} \Phi_{ij}\theta_j\right)\right).$$

Simplifying, we obtain

$$g(\theta,\nu^*(\theta)) = -\sum_{j=1}^{k} \theta_j \mu_j - \ln\left(\sum_{i=1}^{n} \exp\left(-\sum_{j=1}^{k} \Phi_{ij}\theta_j\right)\right).$$

Hence, we have written the dual function in terms of θ by maximising over $\nu \in \mathbb{R}^{n+1}$ with $\nu_i \geq 0$ for $i = 1, 2, \ldots, n$.

Role of θ^* in p^*:
In the optimal solution p^*, the value of p_i^* is determined by the direction of θ^* with respect to Φ_i.

$$p_i^* \;\propto\; \exp\!\left(-(\theta^*)^\top \Phi_i\right),$$

and $(\theta^*)^\top \Phi_i$ depends on the direction of θ^* with respect to Φ_i.

(e) Library used for solving primal problem – SciPy

Result:

$$p^*_{\text{primal}} = \begin{bmatrix} 0.10321036 \\ 0.12905792 \\ 0.13714048 \\ 0.13406610 \\ 0.11262183 \\ 0.08791059 \\ 0.07328297 \\ 0.07473030 \\ 0.06631677 \\ 0.08166269 \end{bmatrix} \tag{60}$$

$$H(p^*) = 2.2700535916544426$$

Plots:

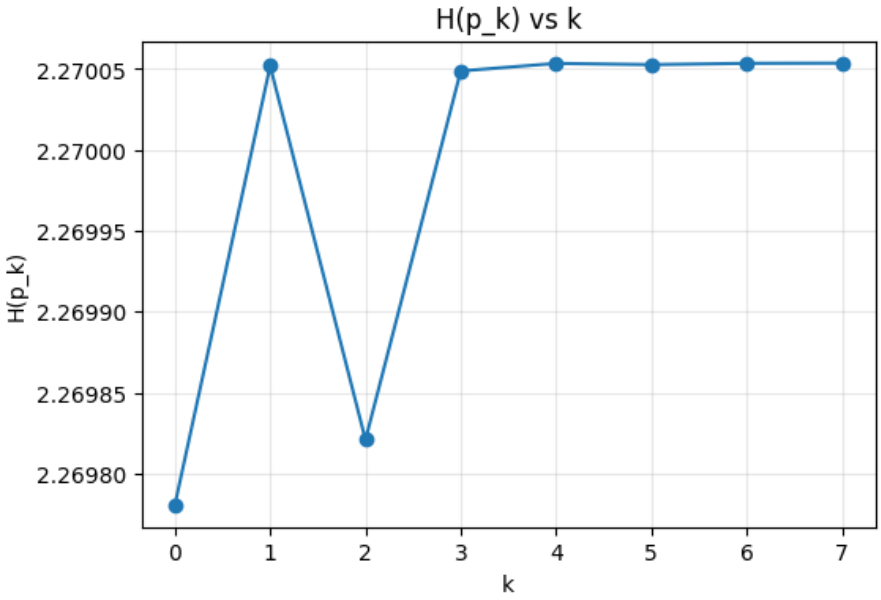


Figure 1: $H(p_k)$ vs. iteration k.

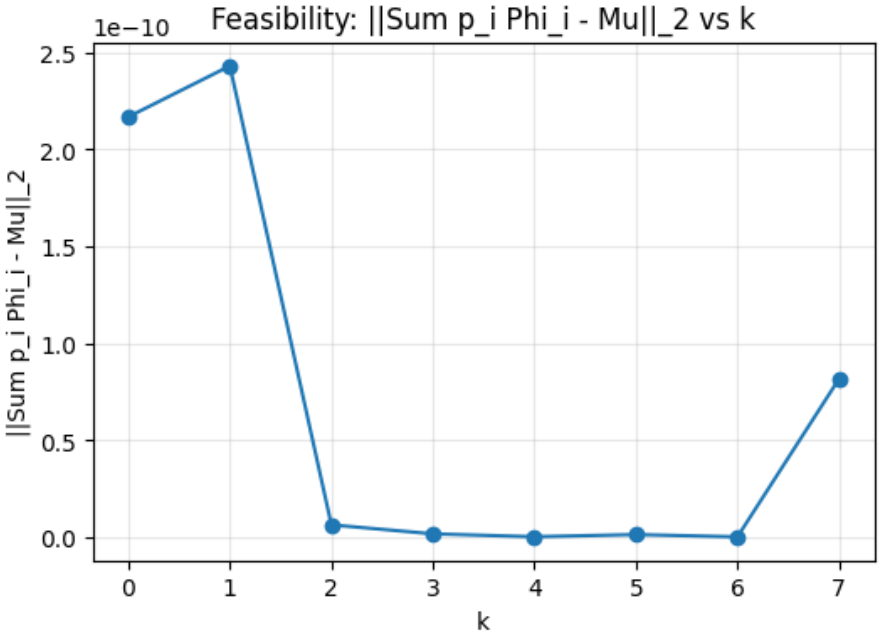


Figure 2: Constraint residual $\|\Phi^\top p_k - \mu\|_2$ vs. iteration k.

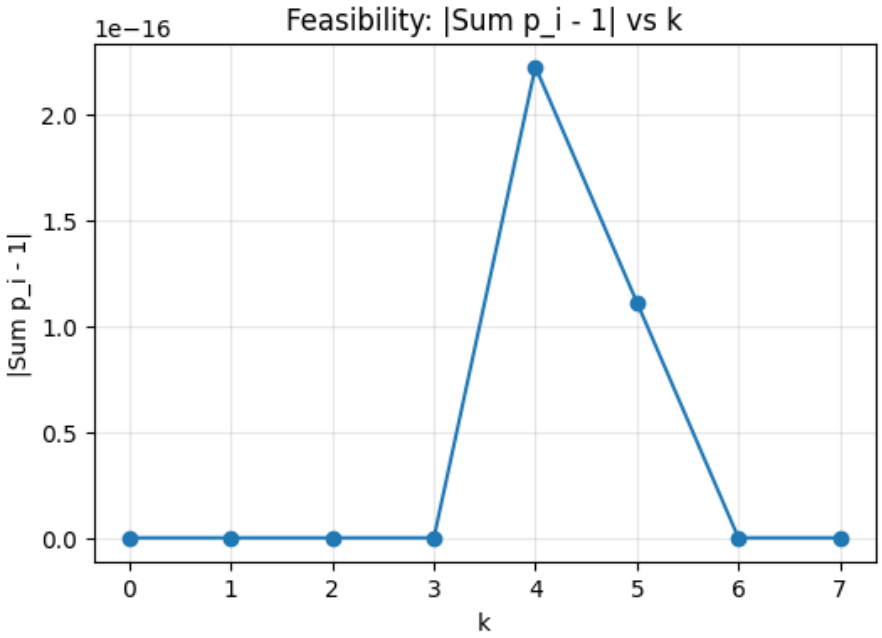


Figure 3: Simplex residual $|1^\top p_k - 1|$ vs. iteration k.

(f) We now move on to solve the dual problem described by equations 15 and 16 using CVXPY.

Results:

$$\theta^* = \begin{bmatrix} -0.06253605 \\ -0.32716869 \end{bmatrix} \tag{61}$$

$$\nu^* = \begin{bmatrix} 8.45884997 \times 10^{-9} \\ 6.10144317 \times 10^{-9} \\ 5.48079000 \times 10^{-9} \\ 5.74560042 \times 10^{-9} \\ 7.43747216 \times 10^{-9} \\ 1.10215546 \times 10^{-8} \\ 1.51482680 \times 10^{-8} \\ 1.46225155 \times 10^{-8} \\ 1.80955030 \times 10^{-8} \\ 1.24980450 \times 10^{-8} \\ 1.34833440 \end{bmatrix}$$

$$g(\theta^*, \nu^*) = 2.2700536016542774$$

We now compute the optimal point $\tilde{p}$ which obeys

$$\tilde{p}_i \;\propto\; \exp\!\big(-(\theta^*)^\top \Phi_i\big)$$

as shown in equation 48.

Using θ^* from equation 61 we obtain

$$\tilde{p} = \begin{bmatrix} 0.10321493 \\ 0.12906061 \\ 0.13713778 \\ 0.13406174 \\ 0.11261679 \\ 0.08790820 \\ 0.07328216 \\ 0.07473112 \\ 0.06631958 \\ 0.08166711 \end{bmatrix}$$

The primal problem optimiser p^* is given in equation 60.

Comparing it with $\tilde{p}$ we obtain, by solving the dual problem, we have

$$\| \, p^* - \tilde{p} \, \|_\infty = 5.043138831387073 \times 10^{-6}.$$

The duality gap, or the difference between the optimal values of the primal and dual problems, is given by

$$\big| \, H(p^*) - g(\theta^*, \nu^*) \, \big| = | \, 2.2700535916544426 - 2.2700536016542774 \, | = 9.999834738039226 \times 10^{-9}.$$

Question 2 (Binary Classification)

The process of generating binary or two-class data can be thought of as follows:

1) The data produced depends on a vector $w \in \mathbb{R}^d$, which denotes the vector perpendicular to the hyperplane separating the two classes.

2) For a given $x_i \in \mathbb{R}^d$ and $w \in \mathbb{R}^d$,

$$\Pr(y_i = +1) = \frac{e^{\,w^\top x_i}}{1 + e^{\,w^\top x_i}} = \frac{1}{1 + e^{-\,w^\top x_i}}. \tag{62}$$

Similarly,

$$\Pr(y_i = -1) = \frac{1}{1 + e^{\,w^\top x_i}}. \tag{63}$$

Therefore,

$$\Pr(y_i = y) = \frac{1}{1 + e^{-\,y\,(w^\top x_i)}}, \qquad y \in \{-1, +1\}. \tag{64}$$

When given (x_i, y_i) pairs of data, our goal is to find the w which produced the data.

The w is found by maximising the probability of the data with respect to $w \in \mathbb{R}^d$.

Since the data points can be considered independent, we have

$$\Pr(\text{data}) = \prod_{i=1}^{n} \Pr(x_i, y_i) = \prod_{i=1}^{n} \frac{1}{1 + e^{-y_i (w^\top x_i)}}. \tag{65}$$

To find w corresponding to the data, we must maximise the probability.

This gives us

$$w^* = \arg\max_{w \in \mathbb{R}^d} \prod_{i=1}^{n} \frac{1}{1 + e^{-y_i (w^\top x_i)}}. \tag{66}$$

Since $\log(\cdot)$ is a monotonically increasing function, we can rewrite equation (66) as

$$w^* = \arg\max_{w \in \mathbb{R}^d} \log\left(\prod_{i=1}^{n} \frac{1}{1 + e^{-y_i (w^\top x_i)}} \right). \tag{67}$$

Using the property $\log(ab) = \log a + \log b$, we obtain

$$w^* = \arg\max_{w \in \mathbb{R}^d} \left(-\sum_{i=1}^{n} \log\Big(1 + e^{-y_i (w^\top x_i)}\Big) \right). \tag{68}$$

Equivalently,

$$w^* = \arg\min_{w \in \mathbb{R}^d} \sum_{i=1}^{n} \log\Big(1 + e^{-y_i (w^\top x_i)}\Big). \tag{69}$$

We denote the objective function by $L(w)$:

$$L(w) := \sum_{i=1}^{n} \log\Big(1 + e^{-y_i (w^\top x_i)}\Big). \tag{70}$$

Hence,

$$\min_{w \in \mathbb{R}^d} L(w) \tag{71}$$

corresponds to a probabilistic binary classifier that separates data into two classes.

(b) Visualising the data provided in the csv file :

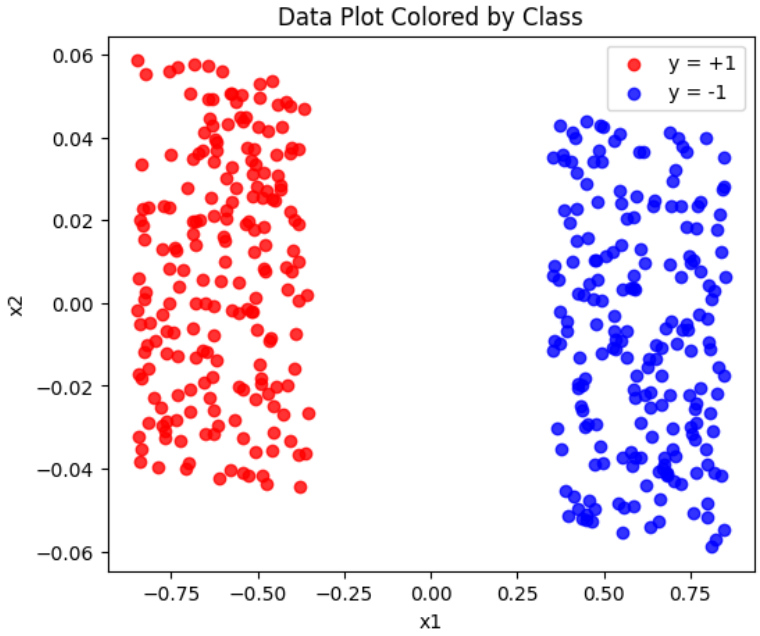


Figure 4: Scatter plot of the dataset coloured by class label y.

From image 4, it is obvious that the data provided is linearly separable. We now move on to minimise the loss for the given data set.

We wish to solve the following optimisation problem for the given dataset

$$\min_{w \in \mathbb{R}^d} \; L(w) \; := \; \sum_{i=1}^{n} \log\left(1 + \exp(-y_i \, x_i^\top w)\right), \tag{72}$$

where $y_i \in \{-1, +1\}$.

Optimiser used to solve the problem in equation 72:

- Optimiser : hand-coded Gradient Descent
- Learning Rate : 0.1
- Initialisation of w : Zero vector
- Number of Iterations : 2×10^4

Plots

1) Plot of Loss function ($L(w_k)$) vs the number of iterations (k).

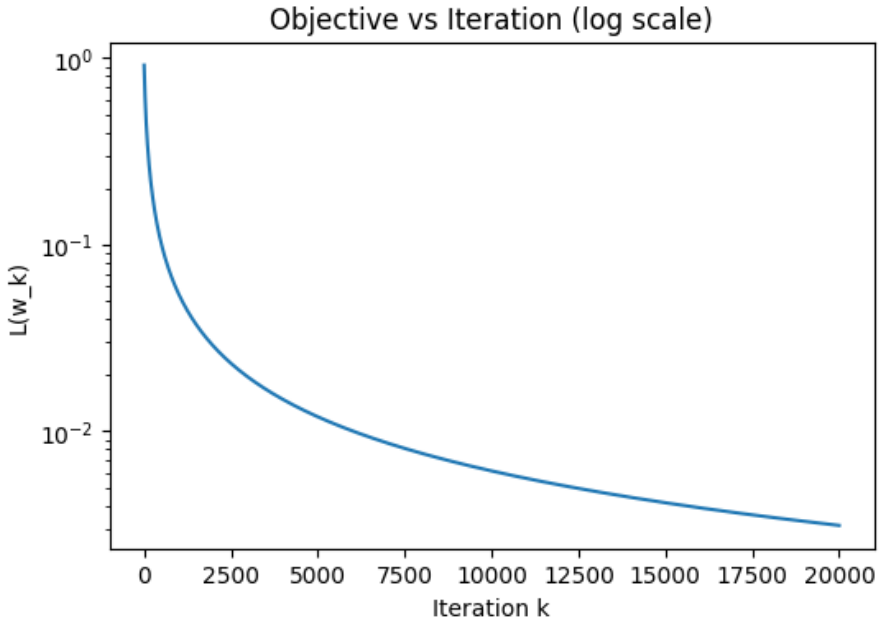


Figure 5: Objective value $L(w_k)$ vs. iteration number k (log scale).

2) Plot of the $L2$ norm of w_k ($||w_k||_2$) vs the number of iterations (k).

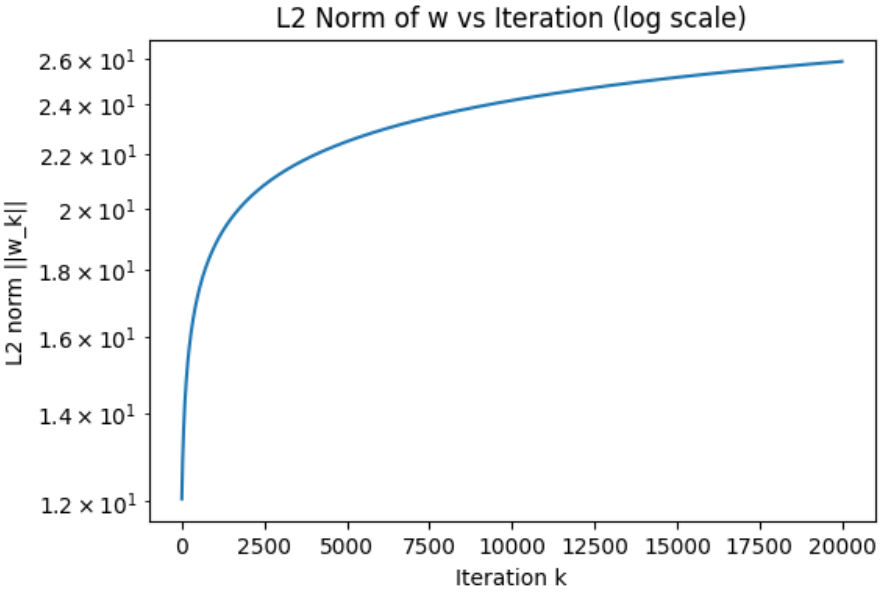


Figure 6: $||w_k||_2$ vs. iteration number k (log scale).

(c) From the plot in Figure 6, it is clear that the norm of the vector w_k increases with the number of iterations k.
Hence, if the gradient descent algorithm is allowed to run for $k \to \infty$ iterations, the norm of w_k will also tend to infinity:

$$\|w_k\|_2 \ \to \ \infty \qquad \text{as } k \to \infty.$$

This failure of the gradient descent algorithm to deliver a finite w_k as $k \to \infty$ is due to the fact that the data is linearly separable.

Since the data is linearly separable, as seen in Figure 4, there exists a vector $w^* \in \mathbb{R}^d$ such that

$$y_i \left((w^*)^\top x_i\right) > 0, \qquad \forall i = 1, 2, \ldots, n. \tag{73}$$

The loss as a function of w^* is given by

$$L(w^*) = \sum_{i=1}^{n} \log\bigl(1 + \exp\bigl(-y_i \, (w^*)^\top x_i\bigr)\bigr) \, . \tag{74}$$

Using (73), the loss becomes

$$L(w^*) = \sum_{i=1}^{n} \log\bigl(1 + \exp\bigl(-\left|(w^*)^\top x_i\right|\bigr)\bigr) \, , \tag{75}$$

(where $|x|$ denotes the absolute value of x).

Replacing w^* by αw^*, where $\alpha > 1$, in equation (75), we get

$$L(\alpha w^*) = \sum_{i=1}^{n} \log\bigl(1 + \exp\bigl(-\alpha \left| \, y_i (w^*)^\top x_i \, \right|\bigr)\bigr) \, . \tag{76}$$

Since we assume strict separability (i.e., $(w^*)^\top x_i \neq 0$ for all i), we have

$$e^{-\alpha \left| y_i (w^*)^\top x_i \right|} \; < \; e^{-\left| y_i (w^*)^\top x_i \right|} \qquad \text{for all } \alpha > 1. \tag{77}$$

we can say that

$$L(\alpha w^*) \; < L(w^*) \qquad \text{for } \alpha > 1. \tag{78}$$

This is what causes the norm of w_k (i.e., $\|w_k\|_2$) to grow in an unbounded fashion.

As $\alpha \to \infty$, we have

$$L(\alpha w^*) \; \to \; 0.$$

Hence, multiplying w^* by a factor greater than one results in a decrease of the loss. Therefore, the gradient descent algorithm fails to deliver a finite optimiser on the dataset when the data is linearly separable.

(d) To arrive at a finite optimiser, we convert the unconstrained optimisation problem defined in 72 into a constrained optimisation problem by placing an upper bound on the L_2 norm of w. In other words, we restrict $\|w\|_2$ to lie below a fixed constant.

Therefore, we modify the optimisation problem as:

$$\min_{w \in \mathbb{R}^d} \; L(w) := \sum_{i=1}^{n} \log\bigl(1 + \exp(-y_i \, x_i^\top w)\bigr) \quad \text{subject to} \quad \|w\|_2^2 \leq k, \tag{79}$$

where $k > 0$ is a constant.
The optimisation problem in (79) is convex because:

1. The constraint set $\{w \in \mathbb{R}^d : \|w\|_2^2 \leq k\}$ is a convex set (a Euclidean ball).
2. The objective function $L(w)$ is convex in w.

Proof of convexity of the objective:

Consider the scalar function

$$f(a) = \log(1 + e^a).$$

Its first derivative is

$$f'(a) = \frac{e^a}{1 + e^a} \geq 0, \qquad \forall a \in \mathbb{R}.$$

Its second derivative is

$$f''(a) = \frac{e^a}{(1 + e^a)^2} \geq 0, \qquad \forall a \in \mathbb{R}.$$

Since $f''(a) \geq 0$, the function f is convex.

Let

$$g_i(w) := f(-y_i\, x_i^\top w), \quad \text{where } w \in \mathbb{R}^d,$$

and $f(a) = \log(1 + e^a)$.

Since $f(a)$ is convex and increasing in a, and $-y_i(x_i^\top w)$ is an affine (and therefore convex) function of w, the composition $g_i(w)$ is convex in w.

Therefore, the objective function

$$L(w) = \sum_{i=1}^{n} g_i(w)$$

is also convex in w, since it is a sum of convex functions.

Hence, we can arrive at the optimiser w^* of the constrained problem in (79) by solving its dual problem, since strong duality holds. The dual problem is given by

$$\sup_{\lambda \geq 0} \inf_{w \in \mathbb{R}^d} \Big(L(w) + \lambda(\|w\|_2^2 - k) \Big). \tag{80}$$

Solving the optimisation problem in 80:

1. We fix $\lambda = 1$.
2. In each step, we minimise over $w \in \mathbb{R}^d$ using `CVXPY`, for a constant λ.
3. We then do gradient ascent w.r.t. λ.
4. The update equation for λ is given by

$$\lambda_{k+1} \;=\; \left[\, \lambda_k + \eta_k \nabla_\lambda \mathcal{L}(w_k, \lambda_k) \,\right]_+ ,$$

   where $[\cdot]_+$ projects onto the constraint set $\lambda \geq 0$.

Results :

1) Plot of the loss function $L(w_k)$ vs. iteration number k (log scale).

2) Plot of the ℓ_2 norm $\|w_k\|_2$ vs. iteration number k (log scale).

3) Final separating hyperplane obtained after convergence.

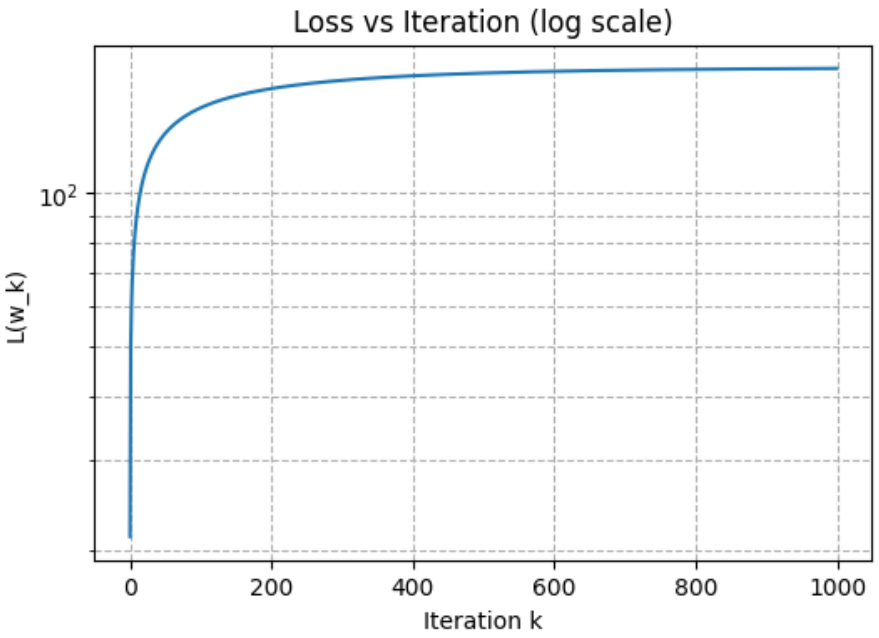


Figure 7: Objective value $L(w_k)$ vs. iteration number k (log scale).

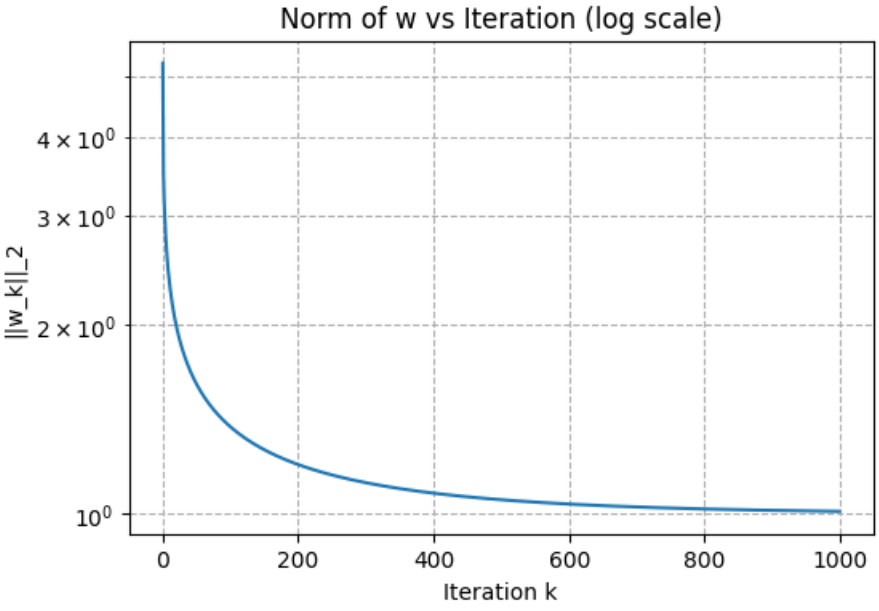


Figure 8: $\|w_k\|_2$ vs. iteration number k (log scale).

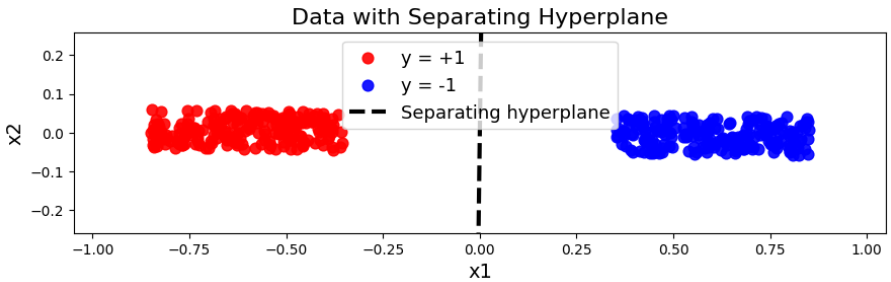


Figure 9: Data plot with the separating hyperplane obtained from the modified method.

The w outputted by our algorithm gives the correct separating hyperplane, suggesting that the algorithm works.

Question 3 (Graph Colouring Problem)

Let $G = (V, E)$ be a simple undirected graph with $|V| = n$. The *vector colouring SDP* to calculate the minimum number of colors required may be formulated as the following optimization problem:

$$\min_{G,\,\rho} \ \rho \tag{81}$$

subject to

$$G \succeq 0, \tag{82}$$

$$G_{vv} = 1 \quad (\forall v \in V), \tag{83}$$

$$G_{uv} \leq \rho \quad (\forall (u, v) \in E), \tag{84}$$

where:

1. $G \in \mathbb{S}^n$ is a Gram matrix, and
2. $\rho \in \mathbb{R}$ bounds edgewise dot products.

The K-color vector feasibility threshold is given by

$$\rho \leq -\frac{1}{K-1}. \tag{85}$$

(a) In this part, we derive the *dual problem* of the vector coloring SDP.
We first construct the Lagrangian with the following dual variables:

1. $\lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ for the equality constraints

$$G_{vv} = 1 \quad (\forall v = 1, 2, \ldots, n), \tag{86}$$

   where $\lambda_i \in \mathbb{R}$ for $i = 1, 2, \ldots, n$.

2. $\alpha_{uv} \geq 0$ for the inequality constraints

$$G_{uv} - \rho \leq 0, \quad \forall (u, v) \in E. \tag{87}$$

   We denote this set of multipliers as α.

3. The dual slack matrix $S \succeq 0$ corresponding to the PSD constraint

$$-G \preceq 0. \tag{88}$$

We make use of the following theorem to incorporate the PSD constraint into the Lagrangian

Theorem 1. *If $A, B \in \mathbb{S}^n$ are both positive semidefinite (PSD), then*

$$\text{Trace}(A^\top B) \geq 0. \tag{89}$$

The Lagrangian for the vector coloring SDP can be written as

$$\mathcal{L}(\rho,\, G,\, \lambda,\, \alpha,\, S) = \rho - \text{Tr}\big(S^\top G\big) + \sum_{(u,v)\in\mathcal{E}} \alpha_{uv}\,\big(G_{uv} - \rho\big) + \sum_{i=1}^{n} \lambda_i\,\big(1 - G_{ii}\big), \tag{90}$$

where the variables α_{uv}, and S satisfy the conditions specified earlier.

The dual function $g(\lambda, \alpha, S)$ is derived from the Lagrangian as follows

$$g(\lambda, \alpha, S) = \inf_{\rho,\, G} \; \mathcal{L}(\rho, G, \lambda, \alpha, S). \tag{91}$$

The dual problem is formulated as

$$\sup_{\substack{\lambda \in \mathbb{R}^n \\ S \succeq 0 \\ \alpha \geq 0}} \; g(\lambda, \alpha, S), \tag{92}$$

where $\alpha \geq 0$ denotes elementwise inequality.

Using equation 91, the dual problem in equation 92 may be written as

$$\sup_{\substack{\lambda \in \mathbb{R}^n \\ S \succeq 0 \\ \alpha \geq 0}} \; \inf_{\rho,\, G} \; \mathcal{L}(\rho, G, \lambda, \alpha, S), \tag{93}$$

Since $\mathcal{L}(\rho, G, \lambda, \alpha, S)$ is linear in ρ and G, to prevent the optimal value from going to $-\infty$, we impose the following conditions.

$$\text{(a)} \quad \nabla_\rho \, \mathcal{L} = 0 \qquad \implies 1 - \sum_{(u,v) \in \mathcal{E}} \alpha_{uv} = 0, \tag{94}$$

$$\text{(b)} \quad \nabla_G \, \mathcal{L} = 0 \qquad \implies -S + M_\alpha - \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) = 0. \tag{95}$$

where:

1. M_α is the matrix whose (u, v)-entry equals α_{uv} if $(u, v) \in E$ $(u < v)$ and all other entries are 0, i.e.

$$(M_\alpha)_{ij} = \begin{cases} \alpha_{uv}, & (i, j) = (u, v), \\ 0, & \text{otherwise.} \end{cases} \tag{96}$$

   As the graph is undirected, M_α is symmetric, which we use to determine $(M_\alpha)_{ij}$ when $j > i$.

2. $\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ denotes the diagonal matrix

$$\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}. \tag{97}$$

The dual problem then reduces to the following:

$$\sup_{\lambda,\, S,\, \alpha} \; \sum_{i=1}^{n} \lambda_i \tag{98}$$

subject to

$$1 - \sum_{(u,v) \in \mathcal{E}} \alpha_{uv} = 0, \tag{99}$$

$$-S + M_\alpha - \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) = 0, \tag{100}$$

$$S \succeq 0, \tag{101}$$

$$\alpha \geq 0. \tag{102}$$

(b) Solving Primal and Dual

The first graph is given by

$$V = \{1, 2, 3, 4\}, \qquad E = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}. \tag{103}$$

Solving the primal optimization problem for this graph, we obtain

$$\rho^\star = -\frac{1}{3}, \tag{104}$$

where $\rho^\star$ denotes the optimal value of the primal problem.

Solving the corresponding dual optimization problem, we obtain

$$d^\star = -\frac{1}{3}, \tag{105}$$

where $d^\star$ denotes the optimal value of the dual problem.

The duality gap is given by

$$\text{Duality Gap} = \rho^\star - d^\star = 0. \tag{106}$$

This was expected as
1) The primal problem is a convex optimization problem.
2) There exist G' and ρ' such that

$$G'_{vv} > 0 \quad \forall\, v \in V, \tag{107}$$

$$G'_{uv} = 1 \quad \forall\, (u, v) \in V, \tag{108}$$

$$G'_{uv} < \rho' \quad \forall\, (u, v) \in E. \tag{109}$$

3) The optimal value of the dual problem satisfies

$$d^\star > -\infty. \tag{110}$$

As the conditions (115)–(118) are satisfied, strong duality holds.

The second graph is

$$V = \{1, 2, 3, 4, 5\}, \qquad E = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 1)\}. \tag{111}$$

Solving the primal problem for this graph, we obtain

$$\rho^\star = -0.81. \tag{112}$$

Solving the dual problem for this graph, we obtain

$$d^\star = -0.81. \tag{113}$$

The duality gap is given by

$$\text{Duality Gap} = \rho^\star - d^\star = 0. \tag{114}$$

This was expected due to the same reasons that were listed earlier.

(c) Feasibility of 3 colorability

(a) Strong duality holds in the Graph Colouring problem as

1) The primal problem is a convex optimization problem.
2) There exist G' and ρ' such that

$$G'_{vv} > 0 \quad \forall\, v \in V, \tag{115}$$

$$G'_{uv} = 1 \quad \forall\, (u, v) \in V, \tag{116}$$

$$G'_{uv} < \rho' \quad \forall\, (u, v) \in E. \tag{117}$$

3) The optimal value of the dual problem satisfies

$$d^\star > -\infty. \tag{118}$$

(this is given).

Hence, we have

$$d^\star = \rho^\star, \tag{119}$$

where $\rho^\star$ denotes the optimal value of the primal problem and $d^\star$ denotes the optimal value of the dual problem.

It is given that for the dual problem,

$$\sum_{i=1}^{n} \lambda_i > -\frac{1}{2}. \tag{120}$$

which implies

$$d^\star > -\frac{1}{2}. \tag{121}$$

As indicated by equation (98), the objective function for the dual problem is given by $\sum_{i=1}^{n} \lambda_i$.
Therefore, using equation (119) and te dual problem defined in equation 98, we can say that

$$\rho^\star > -\frac{1}{2}. \tag{122}$$

For the K-coloring of the graph to be feasible, we require

$$\rho^\star \leq \frac{1}{k-1}. \tag{123}$$

For $k = 3$, equation (123) becomes

$$\rho^\star \leq -\frac{1}{2}. \tag{124}$$

However, (122) does not satisfy (124).

Therefore, we can say that if the dual optimum satisfies (120), then the vector 3-coloring relaxation is infeasible.

(b) For graph 1:

We have

$$\rho^\star = -\frac{1}{3}. \tag{125}$$

For the graph to be 3-colourable, we require

$$\rho^\star \leq \frac{-1}{3-1}, \tag{126}$$

which reduces to

$$\rho^\star \leq -\frac{1}{2}. \tag{127}$$

Since equation 125 does not satisfy the required condition in 127, the graph is not 3-colourable.

For graph 2:

We have

$$\rho^\star = -\frac{1}{1.249}. \tag{128}$$

Since the above equation does satisfy the required condition in equation 127, the graph is 3-colourable.